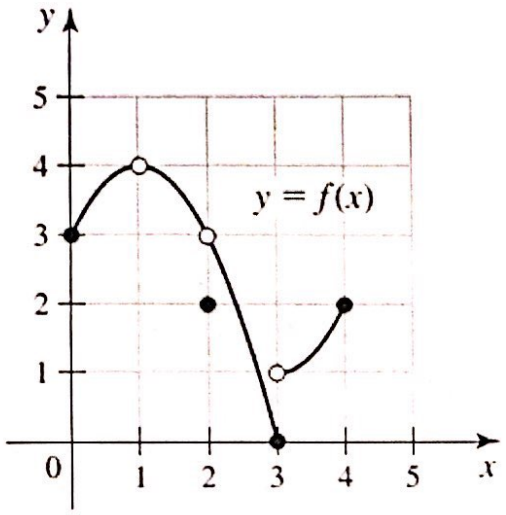
Name: Kanageshwaran

Math 400 Exam 1 Part 4 30 Total Points

(7 pts) 1. You are given the graph of a function f. Find the following limits, if they exist. If the limit does not exist, then explain why using **appropriate limit notation**.



$(a.)$ $\lim_{x\to 2} f(x)$

$\lim_{x\to 2} f(x) = 2$

$(b.)$ $\lim_{x\to 3} f(x)$

$\lim_{x\to 3} f(x) = 0$

$(c.)$ $\lim_{x\to 0^+} f(x)$

$\lim_{x\to 0^+} f(x) = 3$

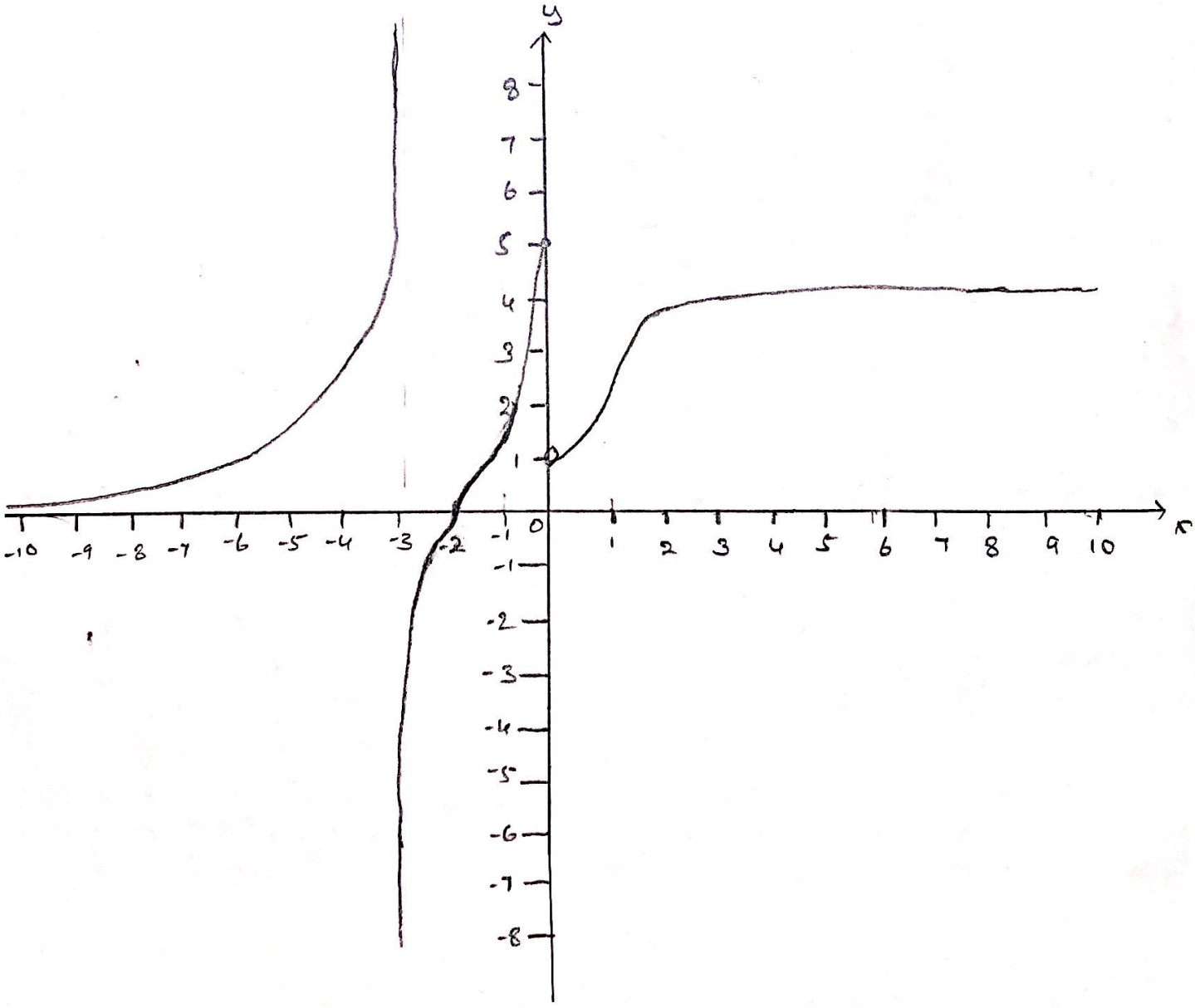
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(6 pts) 2. Sketch the graph of an example of a function f that satisfies all of the given conditions. Include dashed lines for any asymptotes. Include the equations of any asymptotes. Answers may vary.

$f(0) = 5$ $\quad \lim_{x \to 0^-} f(x) = 5$ $\quad \lim_{x \to 0^+} f(x) = 1$

$\lim_{x \to -\infty} f(x) = 0$ $\quad \lim_{x \to -3^-} f(x) = \infty$ $\quad \lim_{x \to -3^+} f(x) = -\infty$

$\lim_{x \to \infty} f(x) = 4$



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(7 pts) 3. Find the following limit **algebraically**. I want the exact limit, not a decimal approximation.

$\lim_{x\to\infty}\left(11x-\sqrt{121x^2+2x}\right)$

$$= \lim_{x\to\infty}\left(11x-\sqrt{121x^2+2x}\right)\left(\frac{11x+\sqrt{121x^2+2x}}{11x+\sqrt{121x^2+2x}}\right)$$

$$= \lim_{x\to\infty}\frac{(11x)^2-\left(\sqrt{121x^2+2x}\right)^2}{11x+\sqrt{121x^2+2x}}$$

$$= \lim_{x\to\infty}\frac{121x^2-121x^2-2x}{11x+\sqrt{121x^2+2x}}\times\left(\frac{1/\sqrt{x^2}}{1/\sqrt{x^2}}\right)$$

$$= \lim_{x\to\infty}\frac{-2x}{11x+\sqrt{121x^2+2x}}\times\left(\frac{1/\sqrt{x^2}}{1/\sqrt{x^2}}\right)$$

$$= \lim_{x\to\infty}\frac{-\frac{2x}{\sqrt{x^2}}}{\frac{11x}{\sqrt{x^2}}+\sqrt{\frac{121x^2}{x^2}+\frac{2x}{x^2}}}$$

$$= \lim_{x\to\infty}-\frac{2}{11+\sqrt{121+2/x}}$$

$$= -\frac{2}{11+\sqrt{121}} = -\frac{2}{11+11} = -\frac{2}{22}$$

$$\lim_{x\to\infty}\left(11x-\frac{1}{11}\sqrt{121x^2+2x}\right) = -\frac{1}{11}$$

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2 Precise Definition of a Limit Let f be a function defined on some open interval that contains the number a, except possibly at a itself. Then we say that the **limit of $f(x)$ as x approaches a is L**, and we write

$$\lim_{x \to a} f(x) = L$$

if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if} \quad 0 < |x - a| < \delta \qquad \text{then} \qquad |f(x) - L| < \varepsilon$$

(5 pts) 4. Prove the statement by using the **Precise Definition of a limit (that is, the ε, δ definition of a limit.)**

$\lim_{x \to 7}(10x + 3) = 73$

Let $f(x) = (10x + 3)$

$|f(x) - L| < \varepsilon_1 = |((10x + 3) - 73)| < \varepsilon_1 = |10x - 70| < \varepsilon_1$

$= |10|\,|x - 7| < \varepsilon_1 = |x - 7| < \varepsilon_1/10$

Choose $\boxed{\delta = \varepsilon_1/10}$

Suppose $0 < |x - 3| < \delta$

Let $\varepsilon > 0$

Then $|f(x) - L| = |(10x + 3) - 73| = |10(x - 7)|$

$= 10|x - 7| < 10\,\varepsilon_1/10 = \varepsilon_1$

Therefore, $\boxed{\lim_{x \to 7} (10x + 3) = 73.}$

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(5 pts) 5. Find the following limit **analytically**. If the limit is ∞ or $-\infty$, then state this.

$$\lim_{x\to\infty} 2^{\csc^{-1}(\ln x)}$$

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